

ANU Physics Academic Integrity Declaration Form

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Date: 4 oct 2026 Assessment Item: assignment 7

Collaboration with others

You are encouraged to discuss problems with other students, however you must submit your own work. The minimum penalty for plagiarism is a fail on the assessment task, and all incidents of plagiarism are recorded on a student's academic record.

If you have collaborated with others, please note these people here:

Use of AI Tools

You may not present material produced by generative AI as your own work, as this is an academic integrity breach. If you have used AI in this assessment, the use must be declared here:

☒ **No AI tools were used** for this assignment

☐ **AI tools were used** as described below:

AI Tools Used: _____

How AI was used (check all that apply):

☐ Mathematical calculations ☐ Problem solving ☐ Code generation
☐ Writing assistance ☐ Research ☐ Other: _____

Chat Evidence Required - you must include your complete AI conversation(s):

☐ Full chat transcript pasted below
☐ Full chat screenshots/files attached
☐ Shared link: _____

I declare that:

- ☒ This AI use declaration is complete and accurate
- ☒ I have included unedited records of all AI interactions
- ☒ I understand the physics concepts and can explain my work
- ☒ I remain responsible for all content in my submission

Signature:  Date: 4 oct 2026

Assignment 7

October 4, 2026

Question 1

(a)

we are given:

- intensity of the light: $I = 1 \text{ W m}^{-2}$
- binding energy: $E_{\text{bind}} = 1.8 \text{ eV}$
- Bohr radius: $r = 5.3 \times 10^{-11} \text{ m}$

we assume that the atom absorbs all the radiation that falls within its Bohr radius.

the area of the atom is approximately the area of a circle,

$$A = \pi r^2$$

substituting,

$$A = \pi(5.3 \times 10^{-11})^2$$

$$A \approx 8.82 \times 10^{-21} \text{ m}^2$$

the intensity is power per unit area,

$$I = \frac{P}{A}$$

so,

$$P = IA$$

substituting,

$$P = (1)(8.82 \times 10^{-21})$$

$$P = 8.82 \times 10^{-21} \text{ W}$$

the binding energy is given in eV, so we convert it to joules.

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

therefore,

$$E_{\text{bind}} = 1.8(1.602 \times 10^{-19})$$

$$E_{\text{bind}} \approx 2.884 \times 10^{-19} \text{ J}$$

using

$$E = Pt$$

$$t = \frac{E}{P}$$

substituting,

$$t = \frac{2.884 \times 10^{-19}}{8.82 \times 10^{-21}}$$

$$t \approx 32.7 \text{ s}$$

hence,

$$\boxed{t \approx 33 \text{ s}}$$

so, it would take about 33 seconds to collect enough energy to eject an electron.

(b)

the calculation in part (a) treats light as a continuous wave, so the electron slowly builds up energy until it has enough to escape.

in reality, light comes in packets called photons, each with energy $E = hf$. an electron absorbs one whole photon at a time. if one photon has enough energy ($hf \geq \phi$), the electron is ejected immediately.

making the light weaker only means fewer photons arrive, not that each photon has less energy. so emission still starts instantly.

Question 2

(a)

we have light travelling from a medium with refractive index

$$n_1(\lambda)$$

into a medium with

$$n_2 = 1.$$

using Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

since $n_2 = 1$,

$$n_1 \sin \theta_1 = \sin \theta_2$$

therefore,

$$\theta_2 = \arcsin(n_1 \sin \theta_1)$$

we want to find how much the refracted angle changes when the refractive index changes.

differentiate with respect to n_1 ,

$$\boxed{\frac{d\theta_2}{dn_1} = \frac{\sin \theta_1}{\sqrt{1 - n_1^2 \sin^2 \theta_1}}}$$

as θ_1 increases, the denominator becomes smaller, so the change in θ_2 becomes larger.

however, the light must still be transmitted into the second medium.

from Snell's law,

$$\sin \theta_2 = n_1 \sin \theta_1$$

and since $\sin \theta_2 \leq 1$, we have

$$n_1 \sin \theta_1 \leq 1$$

the maximum possible value occurs when

$$n_1 \sin \theta_1 = 1$$

$$\sin \theta_1 = \frac{1}{n_1}$$

$$\theta_1 = \arcsin\left(\frac{1}{n_1}\right)$$

this is the critical angle.

hence $\boxed{\theta_1 = \theta_c}$ is the best choice of incident angle to maximise the dispersion.

at this angle, $\theta_2 = 90^\circ$

(b)

we are given:

$$n_{400} = 1.8$$

and

$$n_{700} = 1.7$$

the incident angle should be the critical angle for the larger refractive index, since both wavelengths need to be transmitted.

from part (a),

$$\theta_1 = \arcsin\left(\frac{1}{1.8}\right)$$
$$\theta_1 \approx 33.75^\circ$$

- for the 400 nm light,

$$1.8 \sin(33.75^\circ) = \sin \theta_{2,400}$$

since 33.75° is the critical angle,

$$1.8 \sin(33.75^\circ) = 1$$

so,

$$\theta_{2,400} = 90^\circ$$

- for the 700 nm light,

$$1.7 \sin(33.75^\circ) = \sin \theta_{2,700}$$

$$\sin \theta_{2,700} = 1.7(0.5556)$$

$$\sin \theta_{2,700} \approx 0.9444$$

$$\theta_{2,700} = \arcsin(0.9444)$$

$$\theta_{2,700} \approx 70.8^\circ$$

the angle between the two ends of the optical spectrum is therefore

$$\Delta\theta = 90^\circ - 70.8^\circ$$

$$\Delta\theta \approx 19.2^\circ$$

$$\boxed{\Delta\theta \approx 19^\circ}$$

Question 3

the thin lens equation is

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

since the sun is very far away,

$$u \rightarrow \infty$$
$$\frac{1}{u} \rightarrow 0$$

so,

$$\frac{1}{f} = \frac{1}{v}$$
$$\boxed{v = f}$$

so the image of the sun forms one focal length behind the lens.
the sun has an angular diameter of

$$0.5^\circ$$

so the angle from the centre of the sun to its edge is

$$\theta = \frac{0.5^\circ}{2} = 0.25^\circ$$

consider the ray from the centre of the lens to the edge of the solar image.
using the right triangle,

$$\tan \theta = \frac{r}{f}$$
$$r = f \tan \theta$$

substituting $\theta = 0.25^\circ$,

$$r = f \tan(0.25^\circ)$$

the diameter is

$$D = 2r$$

therefore,

$$D = 2f \tan(0.25^\circ)$$
$$D = 0.008726f$$

$$\boxed{D \approx 8.73 \times 10^{-3} f}$$

where D and f are in the same units.